

1. Our Earth and Our Solar System

Heavenly Bodies

The sun, the moon, the stars, the planets, etc. are all known as heavenly bodies.

Stars

- The heavenly bodies that twinkle are called stars.
- Stars have their own light.
- The sun is also a star. It is closer to us than any of the other stars. Hence, it appears big and brilliant. In its bright light, during the day, we cannot see other stars.
- Some stars twinkle while some do not.

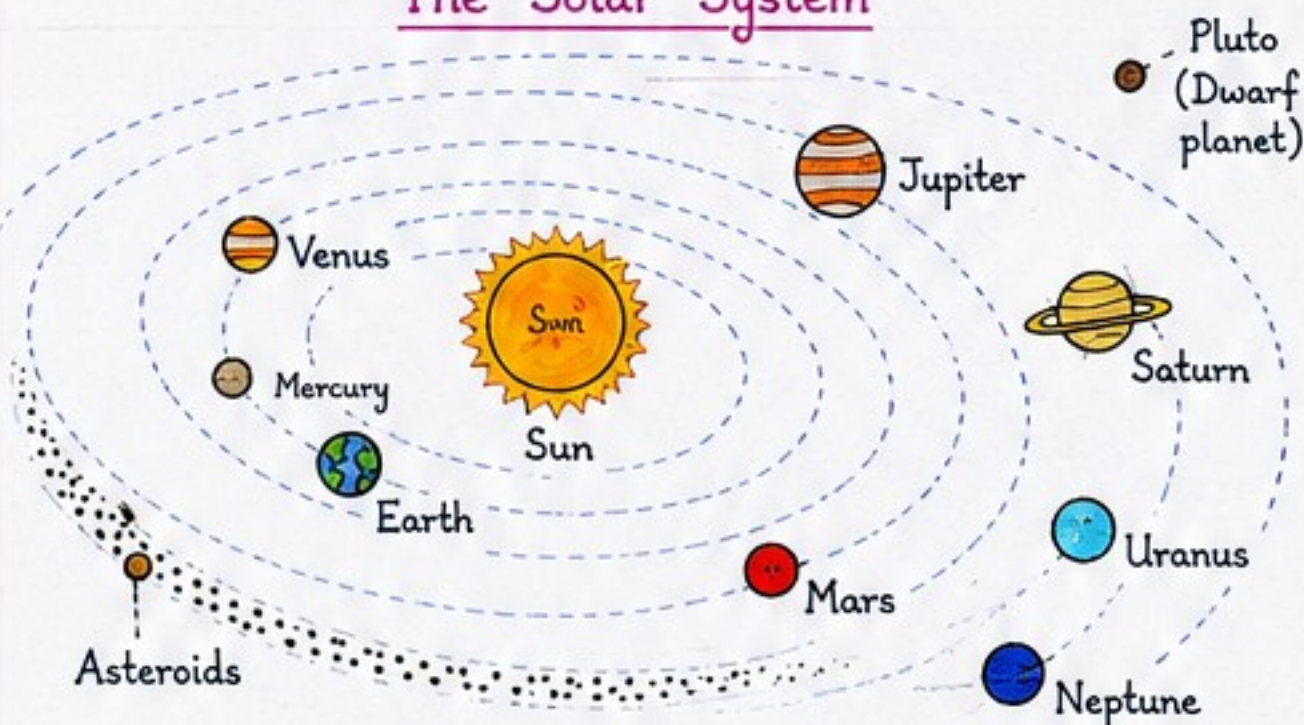
Planets

- The heavenly bodies that do not twinkle are called planets.
- Planets do not have light of their own. They get light from the sun.
- Planets revolve around a star, even as they rotate around themselves.

The Solar System

- Our earth is a planet. It gets its light from the sun.
- It moves around the sun. Its movement around the sun is called the revolution of the earth.
- Besides earth, there are seven other planets that revolve around the sun. They are Mercury, Venus, Mars, Jupiter, Saturn, Uranus and Neptune.
- Every planet in the solar system revolves around the sun along a specific path. This path is known as that planet's orbit.
- The sun, which is a star, and the planets that revolve around it are together called the solar system.

The Solar System



(Only the earth's satellite is shown in it.)

Satellites

- Some heavenly bodies revolve around planets. These are called satellites.
- Satellites too get their light from the sun.
- The moon is a satellite. It revolves around the earth. Hence, it is called a satellite of the earth.
- Most of the planets in the solar system have satellites.

Dwarf Planets

- There are some smaller heavenly bodies that revolve around the sun. They are called dwarf planets.
- Of these, the most well-known is Pluto.
- Dwarf planets revolve independently around the sun. They have an orbit of their own.

Asteroids

- Between the planets Mars and Jupiter, there is a band of numerous small heavenly bodies. They are called asteroids.
- Asteroids also revolve around the sun.

Gravity

- All heavenly bodies exert a force of attraction or a pull on one another. This force is called the force of gravity.
- The sun exerts a gravitational pull on all planets whereas the tendency of the planets is to move away from the sun. As a result of these two forces, a planet keeps revolving around the sun at a fixed distance in a fixed orbit.
- In the same way, satellites revolve around their planets.
- Due to the earth's gravity, all things on the earth remain on it. Even if we throw something upwards with great force, it finally falls down to the ground.

India's Space Missions,

1. **Chandrayaan-1**: On 22 October 2008, ISRO launched a spacecraft to the moon. The mission is known as Chandrayaan-1.
2. **Mangalyaan (M.O.M/ Mars Orbit Mission)**: It was launched on 5 November 2013. It got established in an orbit around the planet Mars on 24 September 2014. ISRO achieved this feat in its first attempt. Both these missions are unmanned, i.e. there were no people on board these spacecraft. The objective of the missions is a deeper study of the moon and Mars.